

LONDON GATWICK, United Kingdom

WMO: 037760

Lat: 51.148N

Lon: 0.190W

Elev: 202

StdP: 14.59

Time Zone: 0.00 (EUW)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	24.8	27.3	20.6	15.6	29.4	23.0	17.5	30.3	26.3	49.0	23.4	48.1	3.0	70	0.548

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.9	81.9	66.4	77.6	64.3	74.9	63.2	67.8	78.1	66.0	75.0	64.4	72.0	7.8	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.4	91.2	71.6	62.7	86.0	69.1	61.1	81.1	67.3	32.2	78.3	30.8	75.2	29.6	71.7	75.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
21.0	18.6	16.5	19.5	87.3	2.9	3.9	17.4	90.1	15.6	92.4	14.0	94.6	11.8	97.4
			19.0	71.1	2.9	1.9	16.9	72.5	15.2	73.6	13.6	74.7	11.5	76.1
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)	
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	51.4	40.8	41.3	44.6	48.9	54.4	59.9	63.9	63.3	58.3	52.6	46.0	41.7	(6)
	DBStd	9.82	6.01	5.73	5.23	5.43	5.24	4.54	4.50	4.31	4.31	5.34	5.89	6.73	(7)
	HDD50	1255	286	247	178	82	16	0	0	0	1	35	145	265	(8)
	HDD65	5067	749	664	634	483	330	164	76	86	205	384	570	722	(9)
	CDD50	1754	2	3	10	49	153	297	431	411	249	116	25	8	(10)
	CDD65	90	0	0	0	0	1	11	42	32	4	0	0	0	(11)
	CDH74	854	0	0	0	5	30	135	374	271	36	3	0	0	(12)
	CDH80	173	0	0	0	0	1	22	83	65	2	0	0	0	(13)
(14) Wind	WSAvg	7.7	8.8	8.7	8.5	7.8	7.8	7.4	7.0	6.6	6.7	7.1	7.2	8.1	(14)
(15) Precipitation	PrecAvg	30.0	3.1	2.0	2.2	2.1	2.0	2.2	1.9	2.2	2.7	3.0	3.2	3.2	(15)
	PrecMax	41.7	7.2	5.6	4.6	4.4	5.6	5.4	4.1	5.2	8.8	7.7	7.7	5.7	(16)
	PrecMin	20.0	0.6	0.3	0.2	0.2	0.0	0.3	0.4	0.0	0.5	0.1	0.6	0.6	(17)
	PrecStd	4.1	1.7	1.3	1.1	1.2	1.1	1.4	1.0	1.3	2.0	1.9	1.8	1.5	(18)
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	55.1	57.3	63.0	73.4	77.3	83.3	86.3	86.2	78.5	69.8	60.8	57.0	(19)
		MCWB	52.4	50.9	52.2	58.6	62.4	66.9	67.7	68.1	64.4	62.6	57.8	54.2	(20)
	2%	DB	53.4	53.5	59.0	66.4	73.4	77.4	82.2	80.8	73.4	64.8	58.7	55.2	(21)
		MCWB	51.0	49.8	51.2	55.1	61.3	63.6	66.4	67.0	63.0	59.3	56.0	53.1	(22)
	5%	DB	51.7	51.7	55.6	62.4	69.5	73.6	78.4	76.6	69.7	62.7	57.0	53.7	(23)
		MCWB	49.5	48.7	49.5	52.9	59.3	62.0	64.8	64.8	60.9	58.4	54.3	51.5	(24)
	10%	DB	50.0	50.0	53.4	58.9	64.7	70.1	74.7	72.0	66.6	60.9	55.2	51.9	(25)
		MCWB	48.4	47.6	48.4	51.1	56.3	60.4	63.3	62.4	59.5	57.4	52.5	49.8	(26)
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	53.2	52.8	54.4	59.6	65.4	68.7	70.1	70.7	67.1	63.2	58.8	54.9	(27)
		MCDB	54.6	56.3	60.3	70.2	74.2	79.8	81.8	81.9	74.8	68.5	60.2	56.2	(28)
	2%	WB	51.7	50.9	52.2	56.1	62.2	65.5	67.5	67.9	64.3	60.8	56.5	53.7	(29)
		MCDB	53.1	52.9	57.0	64.7	70.3	74.8	78.6	77.5	70.7	63.9	58.2	55.1	(30)
	5%	WB	49.9	49.3	51.0	54.0	59.8	63.4	65.8	65.7	62.7	59.5	54.7	52.2	(31)
		MCDB	51.0	50.8	54.6	60.9	67.7	71.0	75.5	73.6	67.4	62.1	56.7	53.8	(32)
	10%	WB	48.3	47.8	49.5	52.0	57.4	61.4	64.1	64.1	61.1	58.1	52.9	50.4	(33)
		MCDB	49.7	49.4	52.3	57.4	64.0	68.6	72.2	70.6	65.3	60.7	55.0	51.8	(34)
(35) Mean Daily Temperature Range	5% DB	MDBR	9.9	11.3	13.7	16.0	16.5	16.9	16.9	16.7	16.0	13.7	11.3	9.9	(35)
		MCDBR	10.1	12.6	18.7	22.3	23.9	24.0	24.4	23.4	20.7	14.6	10.7	8.4	(36)
	5% WB	MCWBR	9.7	9.7	12.2	12.8	13.4	12.4	11.1	11.0	11.5	10.2	9.4	8.3	(37)
		MCWBR	10.1	11.1	15.8	19.4	21.2	20.6	20.8	19.9	16.5	12.6	10.3	8.6	(38)
(40) Clear-Sky Solar Irradiance	taub		0.330	0.345	0.383	0.401	0.406	0.406	0.401	0.393	0.376	0.360	0.333	0.320	(40)
		taud	2.371	2.376	2.267	2.238	2.271	2.296	2.315	2.357	2.397	2.422	2.416	2.361	(41)
	Ebn at Noon		216	246	256	265	269	270	269	265	257	238	213	199	(42)
		Edh at Noon	20	26	35	40	40	40	38	35	30	24	19	17	(43)
(44) All-Sky Solar Radiation	RadAvg		259	476	841	1326	1594	1766	1678	1399	1045	598	310	202	(44)
	RadStd		21	63	109	133	103	163	146	107	81	53	35	25	(45)

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47) Regional (75 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page